

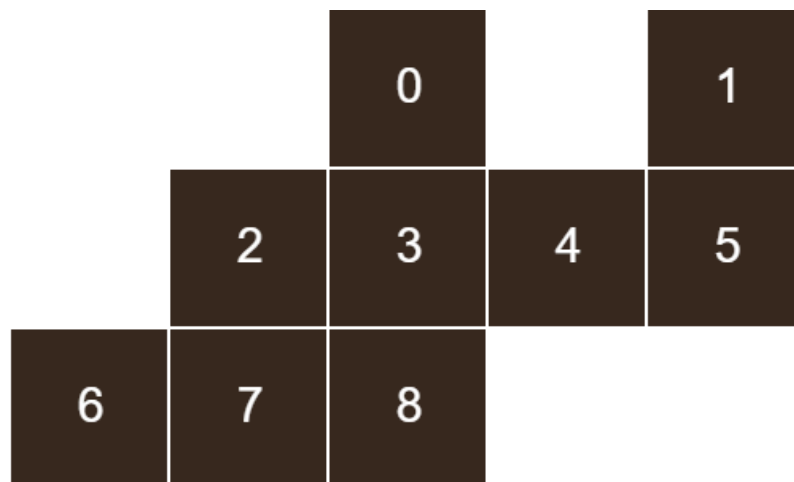


Ancestral Skies Collaboration

Communication 11

Following several years of severe bushfires and storms, large areas of forest and woodland on Indigenous Protected Areas surrounding our telescopes have been significantly damaged. These lands are jointly managed with traditional owners and are home to many culturally and ecologically important native species. We are expanding our collaboration with local communities to provide support in managing these lands.

Rangers are particularly concerned about one region that can be divided into nine management sites, as shown in the map below.



Collectively, these sites support twenty significant native species, including several that are endemic to this region and hold cultural significance for local Indigenous communities. The table below lists the species present at each site.

Site	Species
0	2, 3
1	4
2	2, 16
3	4, 12, 14, 19
4	0, 9, 10, 11
5	1

6	6, 13, 17, 18
7	5, 15, 19
8	1, 6, 7, 8, 13

Ranger groups and ecologists have assessed that all nine sites are in a fragile condition due to habitat loss, invasive species, and changing climate conditions. Due to limited resources, each month conservation teams can focus their efforts on one site, working with traditional owners to restore habitat and cultural land management practices. Restoring a site will permanently secure all the species found there from local extinction.

Assuming each species is of equal ecological and cultural value, which site should the conservation teams prioritise restoring in the first month?

8

Communication 12

Unfortunately, at the end of each month there is a 20% chance that a fragile site will have degraded into a lost state which can no longer be restored, losing all the associated species. Please make use of the function in the following file to work with all the possibilities of fragile sites being lost in a month: [Ancestral3.py](#)

Based on the species data, our current plan is to prioritise restoring the sites in the following order:

8, 6, 4, 3, 7, 2, 0, 5, 1

We will definitely be able to restore Site 8 in the first month but, for example, if Site 6 is lost during the first month then in the second month we will work on restoring Site 4.

If we follow this plan, what is the expected number of species we will save?

14.60

Communication 13

Can you recommend a restoration strategy that would give a higher expected number of species saved? What would be the expected number of species saved?

15.19

Communication 14

In practice, once a site is lost it becomes more likely that neighbouring sites will also be lost. We think the probability of a site being lost in a given month can be quantified as $0.2 + 0.05n$, where n is the number of directly adjacent sites that have already been lost.

Taking this into account, what would be the maximum expected number of species saved now?

14.98

Communication 15

The community has identified that the four species **0, 8, 16 and 19** are the most culturally important. Instead of maximising the expected number of species we save, what should we do if we wanted to maximise the probability of saving all of these species? What would be the probability of being successful?

We look forward to seeing your final presentation.